

THE HONG KONG POLYTECHNIC UNIVERSITY
DEPARTMENT OF COMPUTING
EXAMINATION

Course : Broad Discipline of COMP-61431, BSc Scheme in COMP & AI-61435,
BSc Investment Science & Finance Analytics-63426,
BSc Data Science & Analytics-63425, BSc Scheme in Data Science-63428

Subject : COMP4431 Artificial Intelligence

Group : 1011, 111, 181

Session : 2023 / 2024 Semester I

Date : 20 Dec 2023

Time : 12:30-14:30

Time Allowed: 2 Hours

Subject Lecturer: Dr LIU Yan Fiona

This question paper has ____ 5 ____ pages.
(Some pages may be intentionally omitted.)

Instructions to Candidates:

Answer ALL the questions on the answer book.

Close book.

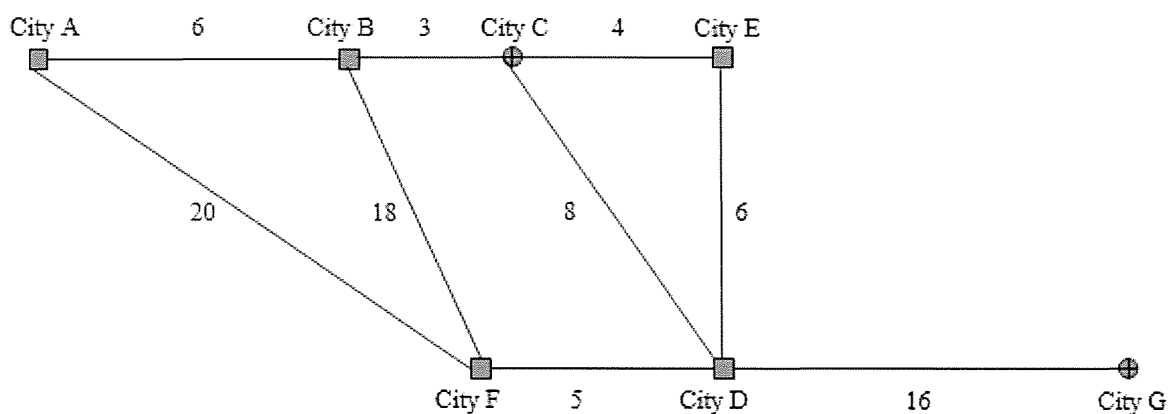
Calculator is allowed.

Do not turn this page until you are told to do so!

Section B**[80 Points]**

Question 3. We want to find a way to drive from City C to City G. We show the distance between two cities in the figure. We formulate the problem to the search tree. The heuristic information is provided as the straight-line distance between two cities as shown below:

A-G: 30; B-G: 25; C-G: 20; D-G: 15; E-G: 10; F-G: 5; G-G: 0



Solve the problem using A* search. Draw the search tree and explain the order of the node checked or/and expanded. **[10]**

Question 4. Clustering algorithm

You are provided with a set of ten one-dimensional data points as follows:

D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
0.12	0.15	0.21	0.50	0.69	0.78	0.89	1.25	1.46	1.97

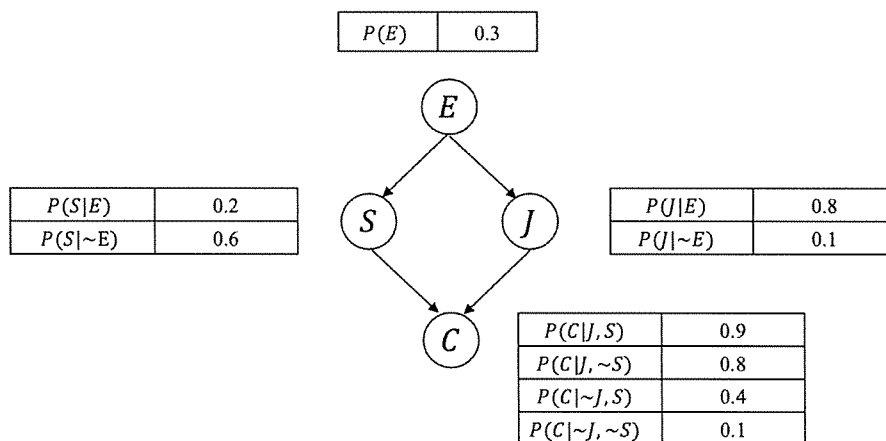
Your task is to cluster these data points into three clusters using the *K*-Means algorithm with the Euclidean distance for clustering. The initial cluster centers are given as:

C1	C2	C3
0.12	0.15	1.97

Please provide a detailed calculation of clustering until the clustering results do not make any change. [12]

Question 5. Bayesian networks

Could you please calculate $P(C | F)$ according to the information given below? [16]

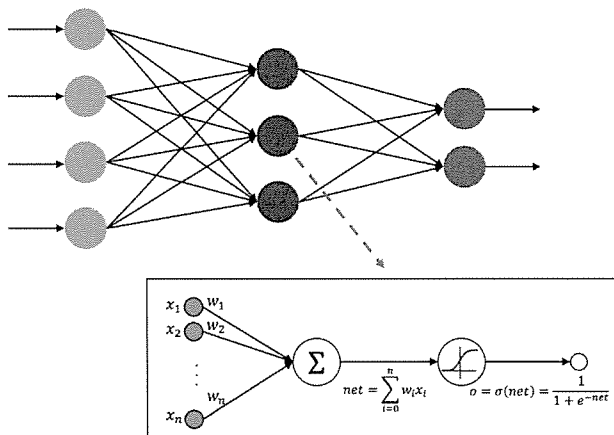
**Question 6. Decision trees.**

Please construct the decision tree to classify the data points with five features into two categories according to the table below. Please write down all the calculations of entropy and information gain. [18]

	F1	F2	F3	F4	F5	Category
D1	2	0	1	2	0	1
D2	0	2	1	3	1	1
D3	1	1	2	3	2	1
D4	2	1	0	1	1	1
D5	1	1	0	2	1	1
D6	2	0	1	0	0	1
D7	2	0	1	0	0	1
D8	0	0	0	3	0	0
D9	2	1	1	1	1	0
D10	0	0	2	1	2	0
D11	1	2	0	3	2	0
D12	1	0	1	3	0	0
D13	2	0	1	1	1	0
D14	2	0	1	2	1	0

Question 7. Neural Networks.

We want to train the following neural network with four input nodes, 3 hidden nodes, and 2 output nodes as shown below.



The output of the neural network can be formulated as:

$$o_j = \frac{1}{1 + e^{-net}} = \frac{1}{1 + e^{-\sum_{i=1}^4 w_{ij}^{(0)} x_i}}$$

$$\tilde{y}_p = \frac{1}{1 + e^{-net}} = \frac{1}{1 + e^{-\sum_{j=1}^3 w_{jp}^{(1)} h_j}}$$

The initial $W^{(0)}[0] = \begin{bmatrix} 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 \end{bmatrix}$ is the weight between the input layer and the hidden layer

while the $W^{(1)}[0] = \begin{bmatrix} 1.0 & 1.0 \\ 1.0 & 1.0 \\ 1.0 & 1.0 \end{bmatrix}$ is the weight between the hidden layer and the output layer. The bias between the input layer and the hidden layer always keeps zero while the bias between the hidden layer and the output layer also keeps zero.

For $W^{(1)}$, $W_{jp}^{(1)}[k+1] = W_{jp}^{(1)}[k] + \eta \delta_p o_j$

where $\delta_p = (y_p - \tilde{y}_p) \tilde{y}_p * (1 - \tilde{y}_p)$

For $W^{(0)}$, $W_{ij}^{(0)}[k+1] = W_{ij}^{(0)}[k] + \eta \delta_j x_i$

where $\delta_j = (\sum_p W_{jp}^{(1)}[k] \delta_p) \tilde{o}_j * (1 - \tilde{o}_j)$

The learning rate η is set to 0.5. Could you please calculate the update $W^{(0)}[1]$ and $W^{(1)}[1]$ with

one input training data $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$?

The number is accurate to 4 decimal places (keep 4 digits after the decimal point) in all calculations. For example, for the number 0.00123456, 0.0012 is used in the calculations and for the number of 1.23456, 1.2345 is used in the calculation. [24]

** END **